TCS Numerical Ability Preparation One Shot | Aptitude Prep for TCS NQT & TCS Ignite Smart ...
ax + by + c = 0
ax² + by + c = 0
TATA
CONSULTANCY
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EQUATIONS
7:49:03 / 11:50:56
TCS Linear Equation
Linear Equation

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Q1. In a competitive exam, 5 marks are awarded for every correct answer and for every wrong answer, 2 marks are deducted. Sathwik scores 32 marks in the examination. If 4 marks had been awarded for each correct answer and 1 mark had been deducted for each incorrect answer, Sathwik would have scored 34 marks. If Sathwik attempted all the questions, how many questions were there in the test ?

A. 14
B. 26 ✓
12
20

Total answers correct = c
Total answers wrong = w
} total questions = c+w

(I) $5c - 2w = 32$ — ①

(II) $4c - w = 34$ — ②

$5c - 2w = 32$
$8c - 2w = 68$

$-5c + 2w = -32$
$8c - 2w = 68$
$3c = 36$

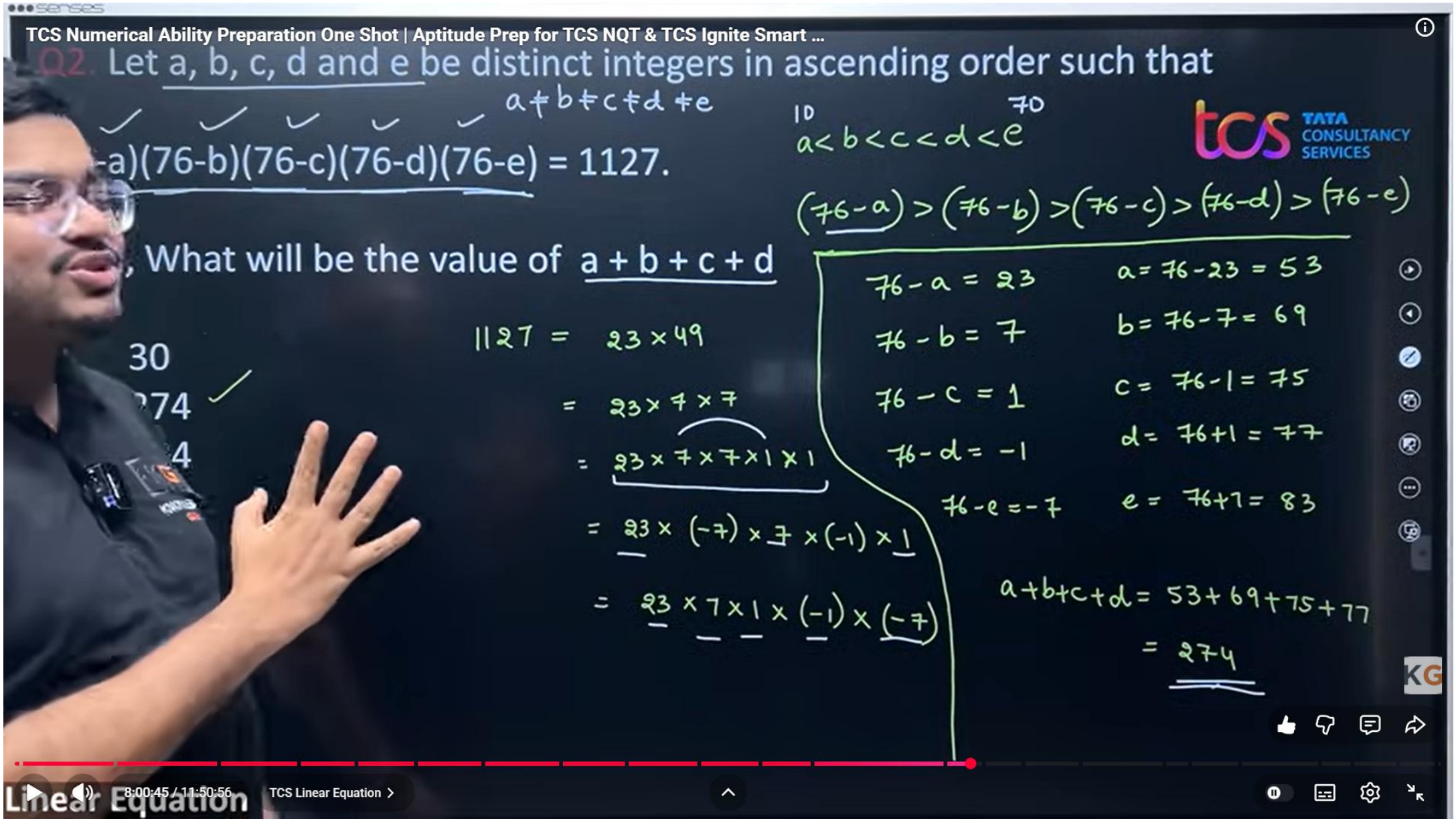
$c = \frac{36}{3} = 12$

$(4 \times 12) - w = 34$

$48 - w = 34 \Rightarrow w = 48 - 34$
$w = 14$

$c + w = 12 + 14$
$= 26$

TATA CONSULTANCY SERVICES
Linear Equation
7:51:04 / 11:50:56
TCS Linear Equation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Q2. Let a, b, c, d and e be distinct integers in ascending order such that
a≠b≠c≠d≠e
-a)(76-b)(76-c)(76-d)(76-e) = 1127.
, What will be the value of a + b + c + d
30
274
4
1127 = 23 × 49
= 23 × 7 × 7
= 23 × 7 × 7 × 1 × 1
= 23 × (-7) × 7 × (-1) × 1
= 23 × 7 × 1 × (-1) × (-7)
10
a < b < c < d < e
70
(76-a) > (76-b) > (76-c) > (76-d) > (76-e)
76 - a = 23
76 - b = 7
76 - c = 1
76 - d = -1
76 - e = -7
a = 76 - 23 = 53
b = 76 - 7 = 69
c = 76 - 1 = 75
d = 76 + 1 = 77
e = 76 + 7 = 83
a+b+c+d = 53+69+75+77
= 274
TATA CONSULTANCY SERVICES
Linear Equation
8:00:45 / 11:50:56
TCS Linear Equation

Q3. Assume that f(1)=0 and f(m+n) = f(m) + f(n) + 4(9mn-1).
For all natural numbers (Integers>0) m and n. What is the value of

A. 5436
B. 4831
C. 5508
D. 4832 ✓

$f(1) \Rightarrow m=1 \quad n=1 \Rightarrow f(2)$

$f(2) \Rightarrow m=2 \quad n=2 \Rightarrow f(4)$

$f(4) \Rightarrow m=4 \quad n=4 \Rightarrow f(8)$

$f(8) \Rightarrow m=8 \quad n=8 \Rightarrow f(16)$

$f(16) \Rightarrow m=16 \quad n=1 \Rightarrow f(17)$ ✓

KG KNOWLEDGE GATE

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Weighted Average

Class 3A ⇒ n_1 students ⇒ Avg marks x_1

class 3B ⇒ n_2 Students ⇒ Avg marks x_2

class 3C ⇒ n_3 Students ⇒ Avg marks x_3

Average marks of class '3' $(\bar{x})$ $= \dfrac{n_1x_1 + n_2x_2 + n_3x_3}{n_1+n_2+n_3}$

10 students → 5 marks

20 students → 10 marks

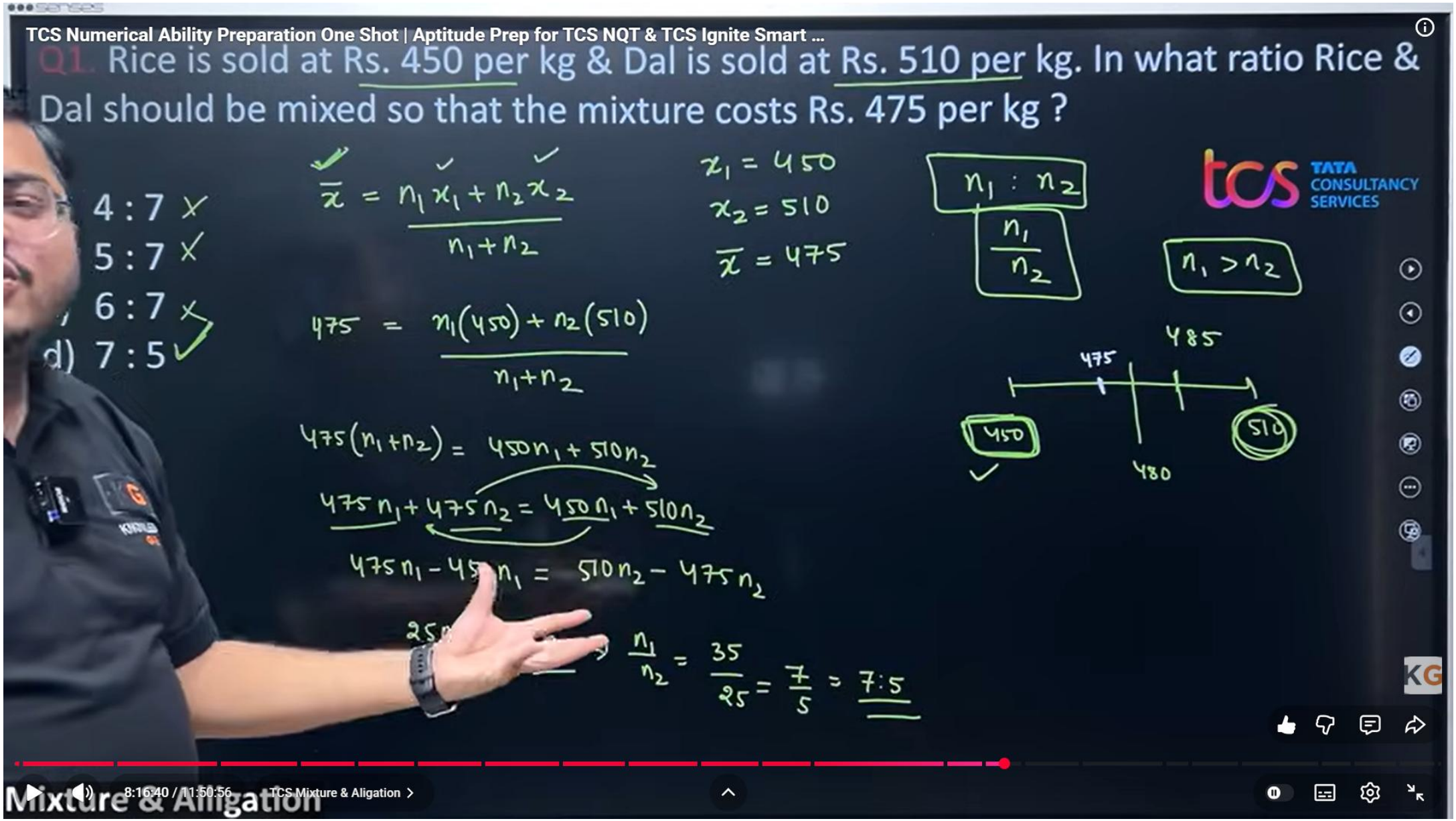
30 students → 20 marks

$$\frac{(10\times5)+(20\times10)+(30\times20)}{10+20+30}$$

$$\frac{50+200+600}{60}$$

$$\frac{850}{60} = \frac{85}{6} = 14.16$$

8:10:23 / 11:50:56 TCS Mixture & Aligation
Mixture & Alligation


TCS Numerical Ability Preparation One Shot | Aptitude Prep for TCS NQT & TCS Ignite Smart ...
Q1. Rice is sold at Rs. 450 per kg & Dal is sold at Rs. 510 per kg. In what ratio Rice & Dal should be mixed so that the mixture costs Rs. 475 per kg ?
4 : 7
5 : 7
6 : 7
d) 7 : 5
$\bar{x} = \frac{n_1 x_1 + n_2 x_2}{n_1 + n_2}$
$x_1 = 450$
$x_2 = 510$
$\bar{x} = 475$
$n_1 : n_2$
$\frac{n_1}{n_2}$
$n_1 > n_2$
$475 = \frac{n_1(450) + n_2(510)}{n_1 + n_2}$
$475(n_1 + n_2) = 450n_1 + 510n_2$
$475n_1 + 475n_2 = 450n_1 + 510n_2$
$475n_1 - 450n_1 = 510n_2 - 475n_2$
$\frac{n_1}{n_2} = \frac{35}{25} = \frac{7}{5} = 7:5$
475
485
450
510
480
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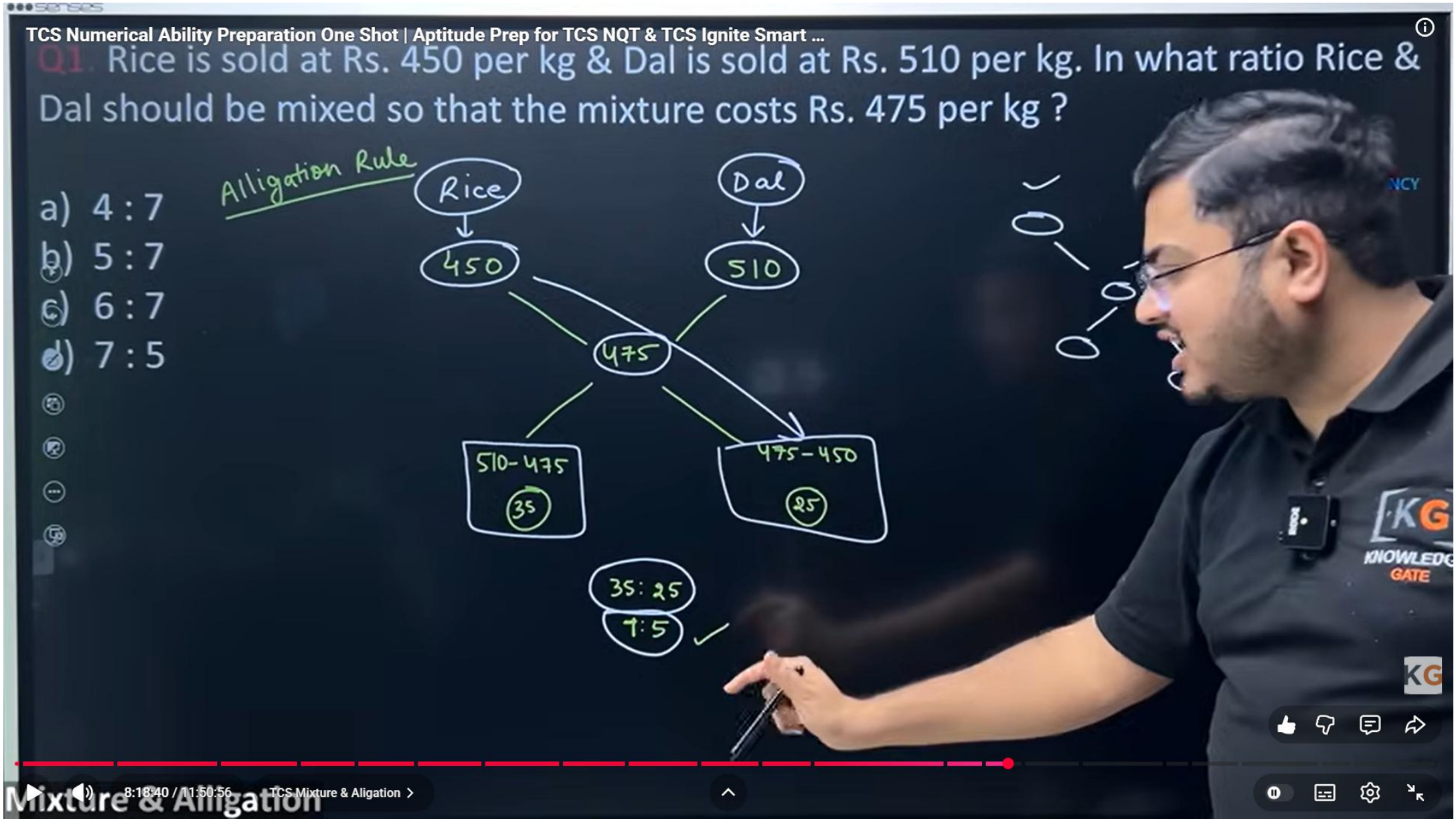
Mixture & Alligation
8:16:40 / 11:50:56
TCS Mixture & Aligation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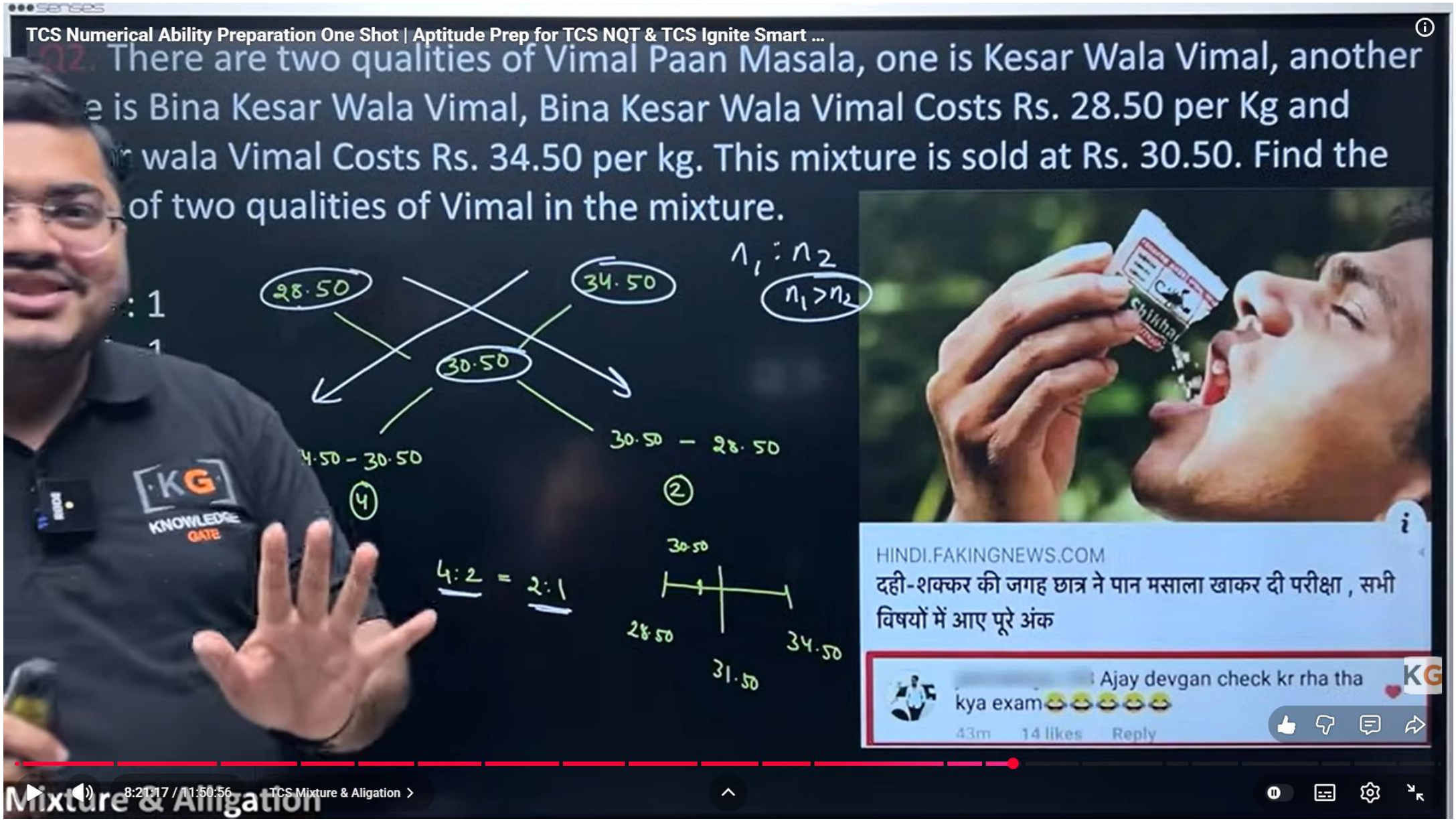
a) 4 : 7
b) 5 : 7
c) 6 : 7
d) 7 : 5
Alligation Rule
Rice
Dal
450
510
475
510-475
35
475-450
25
35 : 25
7 : 5
KG
KNOWLEDGE GATE
8:18:40 / 11:50:56
TCS Mixture & Aligation
Mixture & Alligation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There are two qualities of Vimal Paan Masala, one is Kesar Wala Vimal, another
e is Bina Kesar Wala Vimal, Bina Kesar Wala Vimal Costs Rs. 28.50 per Kg and
wala Vimal Costs Rs. 34.50 per kg. This mixture is sold at Rs. 30.50. Find the
of two qualities of Vimal in the mixture.
28.50
34.50
30.50
4:2 = 2:1
28.50
31.50
34.50
HINDI.FAKINGNEWS.COM
दही-शक्कर की जगह छात्र ने पान मसाला खाकर दी परीक्षा , सभी विषयों में आए पूरे अंक
Ajay devgan check kr rha tha kya exam
43m 14 likes Reply
8:21:17 / 11:50:56
TCS Mixture & Aligation
Mixture & Alligation

Q3. Mrs. Verma has two types of oil, these two types of oil having the rates Rs. 4 per kg and Rs. 5 per kg respectively are mixed in order to produce a mixture having the rate of Rs. 4.6 per kg. What should be the amount of the second type of oil if the amount of the first type of oil in the mixture is 40 Kg?

a) 30 kg
b) 40 k
c)
d) 0 kg ✓

4 5
4.6
5 − 4.6 4.6 − 4
0.4 0.6
0.4 : 0.6
4 : 6

4 : 6
×10 ×10
40 kg : 60 kg

40 : 60
= 4 : 6

4 4.6 5
4.5

Mixture & Alligation